Meridian Health and Wellness Network

Root Cause Analysis

Medication Administration Errors

Client:
Meridian Health and Wellness Network (Fictional)
4800 Peachtree Medical Blvd, Suite 200, Atlanta, GA 30309

Conducted by
Vimala Telu

1.1 Background and Context

Meridian Health and Wellness Network is a fictional multi-facility hospital system operating three acute care hospitals and two outpatient clinics across the greater Atlanta metropolitan area. The network employs approximately 520 registered nurses across all facilities, serving a combined patient census of roughly 1,100 patients per day.

Over a single quarter, the network's Quality and Patient Safety Department recorded 61 verified medication administration errors (MAEs), representing an error rate of 11.2% against the total medication administration events for that period. The industry benchmark for acceptable MAE rates is generally held at below 2% (Institute for Safe Medication Practices, 2022). This significant deviation prompted leadership to commission a formal root cause analysis before determining any corrective action.

It is important to note upfront that not all medication errors result in patient harm. In this instance, 48 of the 61 errors were classified as near-misses or low-severity events that were intercepted before reaching the patient. However, 9 events reached the patient with minor consequences, and 4 were classified as significant events requiring clinical intervention. From a patient safety and risk management perspective, even near-miss events carry serious implications for the organization's accreditation status and liability exposure.

1.2 Performance Gap Statement

Category	Expected Performance	Actual Performance	Gap
MAE Rate	< 2.0%	11.2%	9.2 percentage points above benchmark
Error Interception Rate	≥ 95%	78.7%	16.3 points below expected
Staff Compliance with 5 Rights Protocol	100%	61.3%	38.7% non-compliance observed
Post-Error Documentation Rate	100%	54.1%	Nearly half of errors not documented properly

1.3 Five Whys Analysis

The Five Whys technique was applied in a structured group session involving the Quality and Patient Safety Director, the Clinical Education Manager, and four charge nurses representing each major unit. The session was facilitated in a non-punitive format, emphasizing systemic rather than individual causation.

#	Why?	Finding
1	Why did the medication administration errors occur?	Nurses administered medications without fully completing the 5 Rights verification protocol.
2	Why were nurses skipping the full 5 Rights verification?	Time pressure during peak shift hours, combined with an unfamiliar EHR system interface, caused nurses to rely on memory rather than systematic verification.
3	Why were nurses unfamiliar with the EHR system's medication administration interface?	The hospital had deployed a new EHR system 7 months prior. Initial orientation training lasted only 4 hours and did not include practice scenarios specific to medication administration.

4	Why was the EHR orientation training so limited?	Training was designed and delivered by the EHR vendor rather than clinical education staff. The vendor's standard onboarding did not account for facility-specific workflows or the nursing staff's prior system habits.
5	Why was there no facility-specific supplemental training developed?	The clinical education department was not consulted during the EHR implementation. There was no structured needs analysis or training design plan created before or after go-live.
ROOT CAUSE	Absence of a structured, role-specific, and workflow-integrated training program for the new EHR medication administration module, compounded by inadequate performance support during and after go-live.	

1.4 Fishbone (Ishikawa) Analysis

The fishbone diagram was used to organize the contributing factors identified through interviews, incident report review, and unit observation. Causes are grouped into five major categories, reflecting both system-level and individual-level contributors.

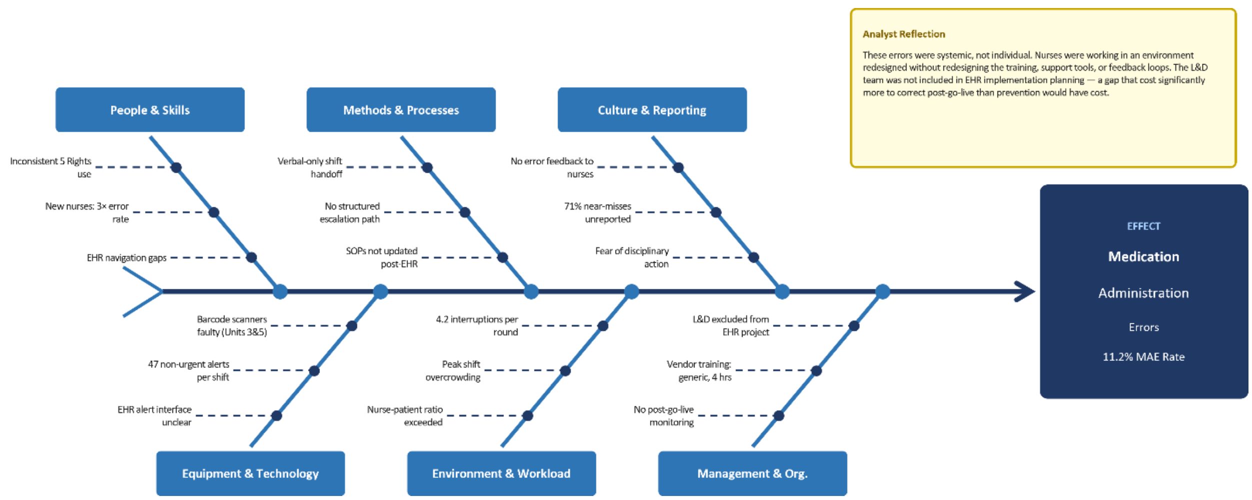

Figure 1.1 — Fishbone (Ishikawa) Diagram: Root Causes of Medication Administration Errors

Meridian Health and Wellness Network | Six-bone analysis across People, Methods, Culture, Equipment, Environment, and Management dimensions

1.5 Gilbert's Behavior Engineering Model (BEM) Application

Gilbert's BEM was applied to determine whether the performance gap is primarily a training problem (knowledge/skills deficit) or an environmental/systems problem. This distinction is critical before recommending any training intervention.

BEM Factor	Environmental (System) Causes	Individual (Performer) Causes
Information	Performance expectations post-EHR not clearly communicated. SOPs not updated to reflect new system workflows.	Nurses unclear on when to escalate alert notifications. Inconsistent awareness of 5 Rights sequences in new EHR.
Resources	Barcode scanners malfunction on 2 of 5 units. EHR alert system generates excessive non-actionable notifications.	Nurses do not have quick-reference job aids at the point of care. No decision-support tools during medication administration.
Incentives	No recognition system for error-free medication rounds. Error reporting is punitive rather than improvement-focused.	No positive reinforcement for protocol adherence. Nurses perceive documentation as administrative rather than safety-critical.
Knowledge & Skills	EHR training was vendor-delivered and not contextualized to clinical workflows.	New nurses lack procedural fluency in the EHR medication module. Experienced nurses apply prior-system habits to the new interface.
Capacity	Peak-hour nurse-to-patient ratios exceed recommended guidelines on 3 units.	Cognitive load during peak hours reduces available attention for verification protocols.
Motives	System does not support error reporting without fear of consequence.	Some experienced nurses express resistance to the new EHR system, preferring the legacy workflow.

1.6 Root Cause Findings Summary

Level	Primary Root Cause	Contributing Factors
Organizational	No L&D involvement in EHR implementation. Training planning absent at project initiation.	No post-go-live performance monitoring framework. Culture of blame suppresses error reporting.
Process	Medication administration SOPs not updated to reflect new EHR workflow steps.	No standardized verification protocol across all units. Inconsistent medication preparation area layout.
Job / Individual	Nurses lack procedural fluency in EHR's medication administration module.	Cognitive overload during peak hours. Alert fatigue. No point-of-care job aids.

The primary root cause is organizational and systemic, not individual. The nurses are not careless — they are working in a system that was redesigned without a corresponding redesign of training, performance support, or feedback mechanisms. Any intervention that addresses only skills without addressing the system will achieve limited and temporary improvement.